

Kavin Sankar

669-245-8975 • kavin.sankar@gmail.com • [linkedin.com/in/kavin-sankar/](https://www.linkedin.com/in/kavin-sankar/) • [Portfolio](#)

EDUCATION

University of Illinois at Urbana-Champaign

Master of Science in Computer Science (4.0 / 4.0)

Champaign, IL

Aug 2025 – Dec 2026

University of Pittsburgh

Bachelor of Science in Computer Science (3.8 / 4.0)

Pittsburgh, PA

Aug 2021 – May 2025

TECHNICAL SKILLS

Languages: Python, C, C++, Java, SQL, MySQL, Postgres, JavaScript, HTML/CSS

Libraries / Software: Apache Kafka, Java Spring Boot, MongoDB, JUnit, Git, Unix/Linux, Jira, BitBucket, Postman

Cloud / Deployment: Docker, Kubernetes, AWS, Jenkins, OpenShift, GCP

EXPERIENCE

Software Engineer Intern

June 2025 – Aug 2025

Capital One

McLean, VA

- Contributed to a system that transformed a **2,500**-human agent manual process into a streamlined, ML-driven platform that reduced transaction dispute resolution times from **60 days to real-time**.
- Developed and deployed a **Python AWS Lambda** function to aggregate fraud decision data from automated services and human agents, integrating with **AWS Step Functions** and other internal Lambdas.
- Wrote comprehensive unit, component, and functional tests using **PyTest** and **Gherkin**.
- Utilized **Docker** for containerization and **Jenkins** for **CI/CD** to ensure high code quality and deployment reliability.

Software Engineer Intern

May 2024 – Aug 2025

PNC Financial Services

Pittsburgh, PA

- Engineered **Apache Kafka** consumer services, ingesting over **125 million** messages daily from users to assist in real-time insight generation regarding customer transactions.
- Implemented deserialization, decryption, and transformation operations on ingested messages.
- Upserted transformed message records, storing data in an **Oracle Database** by using **JDBC** and **SQL**.
- Established CI/CD pipelines, deploying applications on **OpenShift** by using **Docker** and **Jenkins** to scale services.
- Developed applications utilizing **Apache Kafka** and **Java Spring Boot**.

Software Engineer Intern

Sept 2023 – Apr 2024

Pitt Athletics

Pittsburgh, PA

- Developed **3 Python-based microservices** for data ingestion of over **2.4 million** records.
- Integrated data, using **REST APIs** as primary data sources by designing seamless integration workflows.
- Deployed microservices on AWS, utilizing **AWS Lambda** for testing and **AWS Athena** for SQL queries.
- Wrote unit tests, ensuring **90%** code coverage by using **Pytest** for thorough testing.

CS Undergraduate Teaching Assistant

May 2021 – Aug 2021

University of Pittsburgh

Pittsburgh, PA

- Vastly improved students' ability to solve algorithmic questions by providing targeted instruction and practice.
- Taught students multiple data structures such as Linked Lists, Stacks, Queues, & Hashmaps.
- Led **Python** programming lectures, covering core concepts and best practices.
- Taught data analysis tools, covering **Pandas**, **NumPy**, and **Matplotlib** by demonstrating practical applications.

Projects

Cineect

Feb 2024 – Apr 2024

- Developed a movie rec system utilizing a **KNN machine learning** model trained on **33 million** ratings to deliver high-quality movie recommendations.
- Implemented using a **Python Flask API** backend and **JavaScript React** frontend.
- Setup a **CI/CD** pipeline using Google Cloud Build for automated testing and deployment.
- Utilized **Google Cloud Firestore NoSQL** database to store movie information.

Multi-Threaded & Multi-Process Web Server

Oct 2023 – Oct 2023

- Developed two implementations of a web server with **multithreading & multiprocessing**.
- Developed both web servers in **C++**, utilizing threads and worker processes, to improve parallelism.
- Efficiently serves up to **500** requests simultaneously.
- Implemented synchronized logging functionality to record web page requests.

Activities & Leadership

Girls Who Code

Oct 2021 – May 2025

- Introduced over 200 elementary school girls to coding through Scratch and ignited their passion for computer science
- Presented and explained computer science concepts such as the design of algorithms.
- Facilitated discussions between students on real-world applications in computing.

Relevant Coursework:

Distributed Systems, Cloud Computing, Operating Systems, Database Systems, Software Engineering, Machine Learning, Object-Oriented Programming, Parallel Programming, Data Structures & Algorithms